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| <b>AAPG2026</b>                    | <b>CHAAT</b>   | <b>PRC</b> |
| Coordinated by:                    | Uli FAHRENBURG | 48 months  |
| Axe E.01 : Fondements du numérique |                | 672k€      |

## Concurrency and Higher-dimensional Automata: Applications and Theory

**Summary table of persons involved in the project** (p.m. for professors and associate professors is computed assuming 100% research time; non-permanent members in *cursive*)

| Part-ner | Name                       | First name      | Current position    | Role & responsibilities                     | Involvement (p.m.) |
|----------|----------------------------|-----------------|---------------------|---------------------------------------------|--------------------|
| LMF      | <b>Fahrenberg</b>          | Uli             | Professor           | Project coord., member WP1, WP2, WP3, WP4   | 33.6               |
| LMF      | Benjamin                   | Thibaut         | Associate Professor | Member WP2                                  | 9.6                |
| LMF      | Bollig                     | Benedikt        | Research Director   | Member WP1, WP3                             | 7.2                |
| LMF      | Chatain                    | Thomas          | Associate Professor | Member WP1, WP3                             | 12                 |
| LMF      | Faraut                     | Gregory         | Professor (LURPA)   | Member WP3, WP4                             | 9.6                |
| LRE      | <b>Amrane</b>              | Amazigh         | Associate Professor | Partner coord., leader WP3, member WP1, WP4 | 19.2               |
| LRE      | Bazille                    | Hugo            | Associate Professor | Member WP1, WP3, WP4                        | 14.4               |
| LRE      | Laversa                    | Laetitia        | Associate Professor | Member WP3                                  | 9.6                |
| LRE      | Pommellet                  | Adrien          | Associate Professor | Member WP1, WP4                             | 9.6                |
| LRE      | <i>Erlich</i>              | <i>Enzo</i>     | PhD student         | Member WP1                                  | 6                  |
| IRIF     | <b>Ledent</b>              | Jérémy          | Associate Professor | Partner coord., leader WP1, member WP3      | 24                 |
| IRIF     | Clement                    | Emily           | Researcher (LIPN)   | Member WP1, WP3, WP4                        | 9.6                |
| IRIF     | Fortin                     | Marie           | Researcher          | Member WP1, WP3, WP4                        | 9.6                |
| IRIF     | Krivine                    | Jean            | Researcher          | Member WP3                                  | 9.6                |
| LIX      | <b>Dubut</b>               | Jérémy          | Associate Professor | Partner coord., leader WP2, member WP1, WP2 | 19.2               |
| LIX      | Goubault                   | Éric            | Professor           | Member WP2, WP4                             | 9.6                |
| LIX      | Haucourt                   | Emmanuel        | Professor           | Member WP2, WP4                             | 24                 |
| LIX      | Mimram                     | Samuel          | Professor           | Member WP2, WP4                             | 9.6                |
| LIX      | Zeilberger                 | Noam            | Associate Professor | Member WP1, WP2                             | 12                 |
| LIX      | <i>Albert</i>              | <i>Augustin</i> | PhD student         | Member WP2                                  | 12                 |
| LIX      | <i>Chamoun</i>             | <i>Yorgo</i>    | PhD student         | Member WP2                                  | 6                  |
| LIX      | <i>Leclerc</i>             | <i>Louise</i>   | PhD student         | Member WP2                                  | 6                  |
| SMV      | <b>Schlehuber-Caissier</b> | Philipp         | Associate Professor | Partner coord., leader WP4, member WP3      | 32                 |
| SMV      | Dziadek                    | Sven            | Associate Professor | Member WP1, WP3                             | 12                 |
| SMV      | Kushik                     | Natalia         | Associate Professor | Member WP4                                  | 9.6                |
| SMV      | <i>Hajdas</i>              | <i>Jakub</i>    | PhD student         | Member WP1, WP4                             | 24                 |

**Changes compared to the pre-proposal.** The consortium has changed slightly. We have added Emily Clement, researcher at LIPN and co-supervisor of Enzo Erlich, to the IRIF partner, to reinforce the work in WP1 and WP3. We have also added Gregory Faraut, professor at LURPA, to the LMF partner in order to reinforce the work in WP3 and WP4, specifically the new WP4.3 on concurrent control design. Finally, we have added Tarmo Uustalu to the Scientific Advisory Board, in order to help guide the project using his expertise in algebraic and categorical logic and concurrency.

### I. Context, positioning and objective(s) of the proposal

#### a. Objectives and research hypothesis

Higher-dimensional automata (HDAs) are a model for concurrent systems that form a natural extension of standard automata, or state-transition graphs. Despite the richness of the model, HDAs have seen little attention since their introduction in 1991 [97]. Notably, even practical applications such as the translation of Petri nets into HDAs proposed in [112], which should be useful for providing concurrent semantics to Petri nets and aid in their analysis, have had little impact in the concurrency and Petri nets community.

Recently, research on HDAs has greatly accelerated, starting with [45] and initiated by this proposal's coordinator. Significant progress has been made in the field of languages and logics for HDAs: a Kleene theorem which relates HDAs and regular expressions [47]; a Myhill-Nerode theorem which constructs HDAs out of a prefix equivalence on regular languages [49]; a proof of decidability of language inclusion [13]; and notions of monadic second-order logic and first-order logic for HDAs with corresponding Büchi-Elgot-Trakhtenbrot and Kamp theorems [11, 32].

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The research hypothesis of the CHAAT project is that the foundations for a proper *Higher-Dimensional Automata Theory* have thus been laid and should be capitalized upon. Its objective is therefore **to consolidate and expand that theory, develop its extensions to other settings, and apply it in the contexts of model checking and verification.**

### b. Position of the project as it relates to the state of the art

Computing systems which exhibit concurrency, that is the parallel execution of several computations, are notoriously difficult to design, model, and analyze. Indeed, the majority of bugs or errors in modern computing systems are concurrency-related: race conditions, memory violations, etc. The prevention of such errors through formal modeling and analysis is an important topic in theoretical computer science, with applications to many fields from distributed systems to cloud applications.

The majority of formalisms for such analysis employ some version of automata and use an interleaving model of concurrency, in which independence or concurrency of computation steps is modeled by the fact that they may execute in any order: in CCS notation,  $a||b = a.b + b.a$ . As has been recognized long ago notably by the Petri-nets community, this interleaving approach to concurrency poses several problems, or gaps:

1. Concurrency is not modeled faithfully, which may lead to failure of error detection.
2. Models are not well-behaved under *action refinement*, i.e. using variable step granularity.
3. Analysis suffers from state-space explosion due to the high number of interleavings.

On the other hand, HDAs and a few other models, including asynchronous automata and Petri nets, employ non-interleaving concurrency in which the semantics of  $a||b$  is richer, cf. Fig. 1. Nevertheless, and despite their success in modeling concurrent systems, analysis of Petri nets remains a difficult subject; this in itself is a main reason why most analysis formalisms instead use interleaving concurrency.

One particular problem in this regard arises when considering a quantitative notion of time instead of a qualitative one, as is necessary in the modeling of timing constraints. Models such as timed automata have seen some success in analysing industrial-size systems, but they use interleaving semantics and therefore suffer from Problems 1-3 mentioned above. Some timed extensions of Petri nets use non-interleaving semantics, but this induces problems with causality, leading the authors of [31] to opine that “*[t]ime and causality [do] not necessarily blend well in [...] Petri nets*”.

The basis of the CHAAT proposal is the hypothesis that HDAs are a convenient setting in which to address the above problems. They combine the ease of use of an automata-like formalism with the precision of non-interleaving concurrency and lend themselves to integration with other models such as Petri nets or event structures [112]. Notably, regarding time and concurrency, the recent [8, 44] introduce HDTAs, a real-time extension of HDAs, in which time and concurrency appear to blend well.

Through their geometric realization, HDAs also have a strong relation to geometry and topology. HDAs are realized as *directed spaces*, and for the analysis of such spaces, methods and tools from directed algebraic topology may be used [54, 55]. Much work is already available in this area, e.g. [51, 53, 100, 101, 117], and tools from algebraic topology are a frequent source of inspiration for HDA analysis, for example in the use of homotopy for state-space reduction [53] or of mapping cylinders in [47].

Most of the project partners have contributed to the state of the art of HDAs and concurrency theory or directed topology: LMF in [7, 8, 10–13, 21, 22, 31, 42, 44–46, 48, 49, 119], LRE in [7, 8, 10, 11, 13, 14, 96], IRIF in [11, 21, 22, 32, 67, 83], and LIX in [36, 37, 39, 53–55, 63–67]. The SMV partner brings an interest in tool development and applications in the verification of software and communication protocols [12, 14, 27, 40].

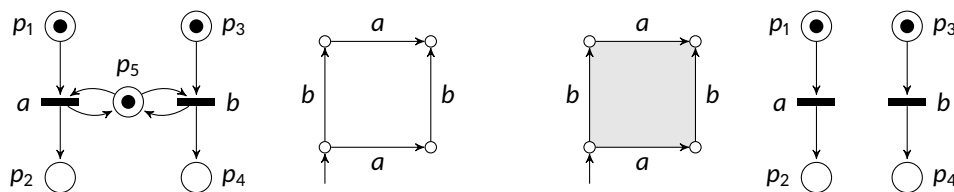


Figure 1: Petri net and HDA models distinguishing interleaving from non-interleaving concurrency. Left: Petri net and HDA models for  $a.b + b.a$ ; right: HDA and Petri net models for  $a||b$ . On the right, the square is filled in, indicating concurrency.

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### c. Methodology and risk management

**Overview.** The CHAAT project is divided into four scientific workpackages (WP1-4) and a management and dissemination workpackage (WP5).

**WP1** is devoted to further develop the theory of *languages and logics* proper to HDAs. Languages of HDAs consist of *pomsets* or partially ordered words, equipped with interfaces and a gluing operation that corresponds to the concatenation of words, see Fig. 2. In WP1 we consider HDAs on finite and infinite pomsets and corresponding logics, notions of bisimulations and corresponding games and modal logics, including action refinement, as well as a framework for active learning of HDAs.

**WP2** aims to push forward the *geometry and topology* of HDAs. We will further develop the homology and homotopy of directed spaces and HDAs, go beyond topology by incorporating quantitative information, and work on the relationship between directed coverings of spaces and unfoldings of computational models.

**WP3** considers variants and extensions of HDAs such as partial HDAs, HDAs with processes or higher-dimensional *timed* automata. We also plan to develop notions of weighted and probabilistic HDAs. We aim to extend the approach of giving a semantics in terms of HDAs to Petri nets by considering additional classes of Petri nets or other concurrency models such as communicating automata through HDAs and their variants. All these developments will be carried out with model checking in mind, as addressed in the following WP4.

**WP4** concerns applications and tools. Its objective is to model and analyse concurrent systems using HDAs, exploiting concurrency information and homotopical structure. We will implement the theory and algorithms developed in the other workpackages in dedicated tools for model checking, translations between models, automata learning, and concurrent control design.

**WP5** gathers the management aspects of CHAAT and the dissemination of its results through a public website, publications, and open-source tools.

**Scientific Advisory Board.** As a complement to the Project Management Committee (see WP5), and given that research on higher-dimensional automata and their applications is essentially an international effort, we have decided to install an international Scientific Advisory Board. The following renowned international experts have kindly agreed to be part of this committee:

- **Henning Basold**, Assistant Professor at Leiden University, The Netherlands, expert in coalgebra and algebraic topology. <https://www.universiteitleiden.nl/en/staffmembers/henning-basold>
- **Christian Johansen**, Professor at the Norwegian University of Science and Technology, expert in concurrency and security. <https://www.ntnu.edu/employees/christian.johansen>
- **Thomas Kahl**, Associate Professor at the University of Minho, Portugal, expert in directed algebraic topology. <https://w3.math.uminho.pt/~kahl/>
- **Georg Struth**, Professor at the University of Sheffield, UK, expert in logical and algebraic methods in concurrency. <https://www.sheffield.ac.uk/cs/people/academic/georg-struth>
- **Tarmo Uustalu**, Professor at Reykjavik University, Iceland, and Lead Research Scientist at the Tallinn University of Technology, Estonia, expert in algebraic and categorical logic and concurrency. <https://cs.ioc.ee/~tarmo/>
- **Rob van Glabbeek**, Professor at the University of Edinburgh, UK, expert in concurrency and operational semantics. <https://www.research.ed.ac.uk/en/persons/rob-van-glabbeek>
- **Krzysztof Ziemiański**, Assistant Professor at the University of Warsaw, Poland, expert in algebraic topology and concurrency. <https://www.mimuw.edu.pl/en/employees/521/>

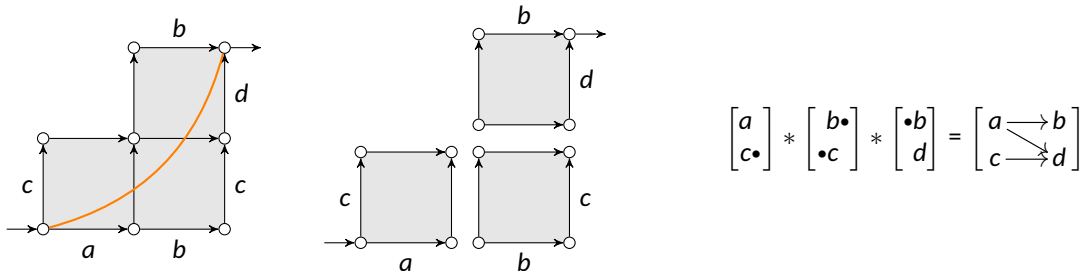


Figure 2: Simple HDA (left) with an execution marked in orange, its decomposition into squares (middle), and (right) the pomset corresponding to the execution, obtained as composition of the pomsets corresponding to the squares (dots denote interfaces).

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The Scientific Advisory Board will advise the Project Management Committee on matters scientific and will help to guide and advance the project's research and to assure its impact.

### **WP1: Languages, bisimulations and logics for higher-dimensional automata**

**Workpackage leader:** Jérémy Ledent (IRIF)

**Participants:** Bollig, Chatain, Fahrenberg, PhD project (LMF). Amrane, Bazille, Pommellet, *Erlich* (LRE). Clement, Fortin, Ledent, Postdoctoral project (IRIF). Dubut (LIX). Dziadek, *Hajdas* (SMV).

**Objectives.** The purpose of this workpackage is to study HDAs in terms of standard properties one addresses for all kinds of automata. The first family of properties falls within the scope of automata as language recognizers with natural questions around expressiveness of the formalism compared to, e.g., logical definability of languages. The second family of properties falls within the context where automata are rather interpreted as system models, and should therefore be subject to an abstraction modulo some equivalence, typically a bisimulation, to capture their behavioral semantics. There, it is natural to seek suitable logics to state and analyze properties of the system behavior, which requires the logical statements to be invariant over equivalence classes.

WP1.1 aims to develop the theory of HDAs as recognizers of finite pomsets. Some main results are already available: a Kleene theorem [47], a Myhill-Nerode theorem [49], a Büchi-Elgot-Trakhtenbrot theorem [11], and a Kamp theorem [32]. Yet the language theory of HDAs is more complicated than the one of standard finite automata, and much remains to be explored. The theory of HDAs as recognizers of *infinite* pomsets ( $\omega$ -HDAs) is much less developed. Some initial results are available in [94], indicating some important differences compared to standard  $\omega$ -automata. The analysis of  $\omega$ -HDA languages is the subject of WP1.2.

The development of suitable bisimulations and other system equivalences, together with corresponding logics, is the subject of WP1.3. We will work on game semantics and action refinement, using the theory of *open maps* [77] as a powerful guiding principle. WP1.4 addresses the issue of synthesizing HDAs through active learning using membership and equivalence queries. Based on recent results on learning pomset automata [96, 114] and the Myhill-Nerode theorem for HDAs [49], we will investigate learning algorithms for HDAs and their applications, with a corresponding implementation in WP4.2.

This workpackage also includes work to consolidate the state of the art on HDAs in a homogeneous and self-contained monograph. The existing literature is scattered across many papers [13, 45–47, 49], with concepts and notations having significantly evolved over the years, which hinders its adoption.

**WP1.1: Language theory of finite HDAs.** The language theory of finite HDAs has seen rapid development in recent years [11, 13, 32, 47, 49]. Yet [13] also shows that HDAs are not determinizable and that their languages are not closed under complement. These are important differences between the automata theory of HDAs and standard automata theory. We intend to pursue several venues in order to further explore these differences and to develop the theory of HDAs proper. One of these is history-determinism, or more generally the spectrum of interesting notions in-between determinism and non-determinism, see [33]. We conjecture that all HDAs are history-determinizable.

Another venue to explore is that of *recognizability*. In standard automata theory, a language is regular iff it is recognized by a finite monoid; yet behaviors of HDAs do not form monoids but rather categories [10]. The proper notion should thus be one of “recognizability by finite categories”, but it is unclear how precisely to define this. Inspiration may be found in recent work by Melliès and Zeilberger [87] as well as in the area of concurrent algebras and monads, for example [72, 74, 106]. Once established, this may also lead to a notion of *aperiodic* languages, important from a logical point of view.

There is also much left to explore regarding relations between languages definable in certain logical formalisms and languages recognised by HDAs. There are several motivations for this. First, it allows us to better understand the expressive power of HDAs. Secondly, translations from logic to automata can be used to turn a model checking problem into one about automata, making connections to WP4.1. A first step in this direction was established in [11], which relates the expressive power of HDAs to that of monadic second-order logic (MSO). On the other hand, our understanding of first-order (FO) definable languages of pomsets is less developed. Studying the class of FO-definable HDA languages is part of Erlich's ongoing PhD. A first step was achieved in [32], where the authors introduce a linear temporal logic (LTL) for HDA languages and show that it is equivalent to first-order logic. Some questions remain open however, as several variants of

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LTL can be considered (e.g., local vs. global LTL), and they should be compared in terms of their expressivity, succinctness, and algorithmic properties. This is connected to WP4.1, model checking HDA properties, as well as WP1.2, since we aim for efficient translations of LTL into  $\omega$ -HDAs.

Some of the work in WP1.1 will be provided by Enzo Erlich, currently a 2nd-year PhD student at LRE supervised by Fahrenberg and Ledent, who will contribute 50% of his time (6p.m.).

**WP1.2: HDAs on infinite pomsets.** The theory of automata on infinite words extends standard automata theory, providing tools and algorithms specifically for modeling non-terminating systems and checking their properties. Together with a Master's student (Luc Passemard), Amrane and Fahrenberg have taken some first steps in the theory of  $\omega$ -HDAs [94]. They studied HDAs with Muller and Büchi acceptance conditions and observed some key differences with the classical theory:

1. Unlike in the finite case, languages of  $\omega$ -HDAs are not closed under order extension (subsumption).
2. Muller acceptance is more expressive than Büchi acceptance, even in the non-deterministic case.

The first difference led to the adaptation of the original rational operations and the introduction of a non-nested  $\omega$ -iteration to define  $\omega$ -rational languages. But, not all  $\omega$ -rational languages are (Muller)  $\omega$ -regular. Thus, many questions remain, particularly in identifying classes of languages where the different definitions coincide. Exploring different types of acceptance conditions inspired by standard  $\omega$ -automata or passing to partial HDAs (WP3.1) may address some of these challenges. In particular, *transition-based* acceptance has shown its force for standard  $\omega$ -automata, and we will explore how to transfer that to  $\omega$ -HDAs. This is part of Jakub Hajdas' ongoing PhD.

A theory of  $\omega$ -HDAs may provide a suitable framework for modeling non-terminating concurrent systems. Moreover, extending the already found connections with logics [11,32] to  $\omega$ -HDAs would be particularly beneficial to WP4.1. We also plan to consider *parity* acceptance conditions and parity *games*, given the important role these play in standard automata theory. Translations from LTL to (deterministic) parity automata and the use of parity games for reactive synthesis are now standard applications of formal methods [75, 102], and we will explore these venues for  $\omega$ -HDAs. This is related to WP4.1 and the focus of the proposed postdoc project at IRIF.

Another important venue is the one of infinite parallelism as used in concurrent Kleene algebra [74] or the  $\pi$ -calculus [90]; [16] shows some preliminary results in this area, and translations from Petri nets (WP3.3), in particular unbounded ones, provide venues which we will explore, for example notions of well-structured systems [56].

**WP1.3: Bisimulations and logics for concurrency and action refinement.** Bisimulation is a classic equivalence relation that determines the behavioral semantics of operational models. While this notion is well mastered for interleaving models since Milner's seminal work on the process algebra CCS [89], the community has faced many attempts to provide a good definition of a bisimulation for non-interleaving concurrency. The expected properties for a good notion of bisimulation are manifold, among which a *game semantics* that potentially gives rise to algorithms to decide the equivalence of models, but also a characterization through an adequate logic [73], and finally congruence properties that allow for compositional reasoning.

In particular, the operation of *action refinement* is very important and has attracted a lot of attention [59, 60, 113] to ensure the expected congruence properties of a good notion of bisimulation for non-interleaving concurrency. Action refinement allows to change the level of abstraction by replacing an action by a model of subactions, thus shifting the granularity of the model description. While there exist several notions of bisimulation for non-interleaving concurrency, most of them fail to be preserved by action refinement, except for variants of *history-preserving* bisimulation [113]. To define action refinement for HDAs, our initial approach will be to introduce a *substitution* operation, as done in [7] for pomset automata; we will then investigate which variants of bisimulation preserve this operation.

For HDAs, some notions of bisimulations have been defined in [38, 42, 112], but their relation to history-preserving bisimulation remains unclear. Safa Zouari is addressing this topic in her ongoing PhD thesis, supervised by Fahrenberg and Ziemiański (Warsaw), with first results available in [119].

The work takes advantage of the open maps framework of [77] which provides a categorical setting of path objects, corresponding bisimulations, and modal logics. [119] is concerned with variants of *ST-bisimulation* and corresponding logics, but provides a good starting point for defining other notions, including history-



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preserving bisimulation, and classifying them together with corresponding logics. We will also relate these to other notions such as MDML [86] or HDML [98]. Through open maps and unfoldings this has relations to WP2.4.

We will explore notions of game semantics for these bisimulations, which in turn opens up for organizing these and other system equivalences into a *linear-time-branching-time spectrum* [111]. Following [43], this may be generalized to a *quantitative* spectrum in order to enable quantitative analysis and verification. We intend to follow this path in order to develop quantitative and robust analysis of HDAs (WP3.2).

**WP1.4: Active learning of models for concurrency.** Active learning consists in inferring a formal model of a black-box system that can be dynamically queried. Interactions with the black-box system are twofold: membership queries and equivalence queries. In contrast, passive learning involves computing a model of a system from a given dataset without the ability to actively query it for additional information. Various automata models can be learnt (DFAs, Mealy machines, tree automata, VPDA's, etc.), but there is a lack of learnable models for concurrency. To our knowledge, the only learning algorithms for concurrent models are those introduced in [23] for learning communicating automata from message sequence charts and in [114] for learning pomset automata. Pommellet and Amrane have recently developed and implemented some active learning techniques in this setting [96] notably inspired by the methods of [107]. In [115], Vinarskii et al. proposed a method for deriving a Timed Finite State Machine (TFSM) from a set of pre-post conditions and further experiments, with a black-box system (implementation). TFSMs with output delays allow modeling concurrency issues such as races between several outputs in response to a given input.

A natural question is to extend active learning techniques to HDAs. Since such techniques often rely on variants of the Myhill-Nerode theorem, and this theorem applies to HDAs [49], it should be possible to design a relevant active learning algorithm. Other concurrency models are also worth exploring. For example, automata over synchronized series-parallel graphs [6] may be particularly relevant, as their underlying theory appears to be well-behaved. Subclasses of communicating automata or HDAs with timers (WP3.2) and extensions to input-output settings will also be of interest. The last point would open up a path to using HDAs in *model-based testing* [2]. To this end, we will investigate the links with distinguishing, homing and synchronization sequences [27, 107], as they are necessary for the integration of learning into model-based testing frameworks. The theoretical results of this workpackage will serve as foundation for tools developed in WP4.2.

Some of the work in WP1.2 and WP1.4 will be provided by Jakub Hajdas, currently a 1st-year PhD student at SMV supervised by Kushik, Dziadek and Fahrenberg, who will contribute 67% of his time (24p.m.).

#### Deliverables and milestones

| ID   | Deliverable/Milestone                                          | Authors | Date               |
|------|----------------------------------------------------------------|---------|--------------------|
| D1.1 | Linear temporal logics and HDAs (PhD thesis E.Erlich)          | IRIF    | T <sub>0</sub> +12 |
| D1.2 | Language theory of finite HDAs                                 | LMF     | T <sub>0</sub> +48 |
| D1.3 | HDAs on infinite pomsets (PhD thesis J.Hajdas)                 | SMV     | T <sub>0</sub> +30 |
| D1.4 | Bisimulations and logics for concurrency and action refinement | LRE     | T <sub>0</sub> +48 |
| D1.5 | Active learning of models for concurrency (PhD thesis)         | LMF     | T <sub>0</sub> +42 |

#### Risks and fallbacks

- **Risk:** In WP1.1, we may find that no good notion of recognizability for HDAs exists. **Risk level:** **medium**. **Impact:** **low**. **Fallback:** Pursue WP1.1 independently of recognizability.
- **Risk:** In WP1.2, we may encounter difficulties with developing good notions of  $\omega$ -HDAs and of translations between logics and  $\omega$ -HDAs, impacting WP4.1. **Risk level:** **low**. **Impact:** **high**. **Fallback:** Re-focus the relevant part of WP4.1 on logics over finite words and translations to HDAs.
- **Risk:** In WP1.3, we may fail to define an HDA analogue of the substitution operation, which is our main venue towards action refinement. **Risk level:** **medium**. **Impact:** **low**. **Fallback:** Other venues are available here, for example using homotopy or subclasses of HDAs, see WP2.4.

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## **WP2: Geometry and topology of higher-dimensional automata.**

**Workpackage leader:** J  r  my Dubut

**Participants:** Benjamin, Fahrenberg (LMF). *Albert, Chamoun, Dubut, Goubault, Haucourt, Leclerc, Mimram, Zeilberger, Postdoctoral project (LIX).*

**Objectives.** Higher-dimensional automata have a realization as geometric objects, obtained by representing each cell as a topological cube and gluing these along faces. In addition, the flow of execution can be represented as a directed structure on this space. This makes HDAs objects of *directed algebraic topology* [54,55] and opens up for the use of methods from topology and geometry in their analysis. Many interesting results are already available in this area, e.g. [51,53,100,101,117], and we will continue to develop geometric and topological methods for the analysis of HDAs.

The work in WP2.1 is concerned with algebraic invariants for directed spaces, viz. homology and homotopy. In standard algebraic topology these are very useful tools for the analysis of topological spaces, and their directed equivalents should play the same role in directed topology. In WP2.2 we will extend the theory of directed topological spaces to incorporate quantitative and metric information, enabling the modeling and analysis of more geometric models such as hybrid systems. In standard topology, homotopy is related to universal coverings, and their directed equivalents are the subject of WP2.3. Directed coverings are related to unfoldings of HDAs and other types of systems, a subject which we intend to explore.

**WP2.1: Algebraic structure of geometric realizations.** Developing good *homology* theories for directed spaces has proven elusive, despite many attempts, including by several members of this project [37,41,63,68,78]. The goal is to find algebraic invariants that classify directed spaces in a meaningful way (geometrically, but also for the concurrency of the underlying system), while having good computability properties (e.g., Mayer-Vietoris sequences, relations to linear algebra). Recently, in [65], Dubut and Goubault's proposal [37] has been connected to persistence in topological data analysis [28] and those constructions have been lifted to general framed bicategories, the convenient setting for modules [3]. Many questions remain: how those homology theories can be efficiently computed using persistence techniques; how they classify directed spaces, even at low dimensions; and how they compare to homology of directed graphs (e.g., [76]). This is the topic of Augustin Albert's ongoing PhD.

Another aspect is *homotopy*, a much finer but uncomputable algebraic invariant of spaces. Traditionally, homotopy is about studying spaces through the lens of higher categories (see [99] for the undirected case, [39] for a directed version). This correspondence is at the core of *Homotopy Type Theory* [110], where thus types are interpreted as  $\infty$ -groupoids, see also [19]. Defining a proper type theory whose models are precisely a variant of  $\infty$ -categories (for example, using variants of cubical sets) is a challenging task, as they fail to be locally cartesian closed: they thus do not form a model of Martin-L  f type theory, but we can hope that they do for a variant which is yet to be defined. A lot is yet to be explored, especially for a proper *directed HoTT* (see [104]). One aspect is to go from the simplicial view of [104] to that of  $\theta$ -spaces [103], allowing to deal with weak higher categories. Some first steps have been taken in [82] and this is the topic of Louise Leclerc's ongoing PhD. A tightly related problem whose solution has eluded the community for a long time is finding a proper language of *model categories* similar to [99] for directed homotopy theory, and for concurrency (see also WP2.3). If some partial answers to both problems can be found in [36,39,47], it is known that the standard language of model categories is inadequate, and further exploration is needed.

Some of the work in WP2.1 will be provided by Augustin Albert and Louise Leclerc, currently 1st-year and 2nd-year PhD students, respectively, at LIX supervised by Goubault, Dubut, and Mimram, who will contribute 50% of their time (12p.m. resp. 6p.m.).

**WP2.2: Geometric models for concurrent hybrid systems.** One motivation underlying the usage of directed spaces is that they allow to model constraints imposed by the physics of the system, for example directed by differential equations or inclusions (see [66]). Yet usual directed spaces and the algebraic methods developed for them leave out crucial aspects of hybrid systems.

The first aspect is quantities, and particularly the metric structure underlying HDAs and hybrid systems. It was recently discovered by Haucourt [71] that this geometry has a lot in common with *chronogeometry*, the underlying geometry of spacetime in relativity [93]. In particular, a well-chosen metric on the geometric

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model  $M$  of a concurrent program ensures that path length corresponds to execution time: as light follows geodesics in spacetime geometry, a “perfect” scheduler should make the instruction pointer follow a geodesic on  $M$ . For the moment, this correspondance concurrency–relativity is limited in scope: it only covers systems whose geometric realization is a product of graphs, and a goal of this task is to extend it to more general classes of systems such as HDAs. Some first steps have been achieved in [29], where a larger class of precubical sets are represented by manifolds, opening up the door to arguments from differential geometry. This is part of the topic of Yorgo Chamoun’s ongoing PhD.

The second aspect is the discrete change of *modes* of hybrid systems. The recent [69] extends the framework of directed spaces to handle “unstoppable” actions, allowing the modeling of instantaneous change of modes in a continuous way. This begs the question of how these extensions affect the algebraic methods developed for homotopy and homology (WP2.1). These also pave the way to geometric foundations of timed and hybrid extensions of HDAs (WP3.2). More generally, this task will consolidate the recent trend of applying topological methods in control problems, such as the synthesis of invariants [58], the complexity of planning problems [66], or the stabilization of trajectories in feedback control [17].

Some of the work in WP2.2 will be provided by Yorgo Chamoun, currently a 2nd-year PhD student at LIX supervised by Haucourt et Mimram, who will contribute 50% of his time (6p.m.).

**WP2.3: Homotopy, unfoldings, and computations.** There is a notion of unfolding of HDAs [36, 42], similar to computation trees for standard transition systems. This has relations to unfoldings of Petri nets (see WP3.3) and also to event structures [91] which we will explore. Another inspiration is topological, in that unfoldings also resemble the universal coverings of algebraic topology. This has led to a theory of dicoverings of directed spaces [50, 52] and hence to a relationship between unfoldings and (directed) homotopy [51]. The precise relation between universal dicoverings and unfoldings of HDAs is not clear, but some hints are in [38]; we intend to work this out.

[38] also points out that unfoldings and universal coverings are tightly related to bisimulations and modal logic, through the lens of open maps [77] (WP1.3). Relations between open maps and unfoldings of *partial* HDAs (WP3.1) can be found in [36], where unfoldings are also interpreted as cofibrant replacements in the language of model categories (WP2.1). There are some works on so-called *tame* (directed) paths [117, 118] which must be taken into account here; tame paths always proceed between points, and every path in a precubical set is homotopic to a tame path. In this context it will be important to explore the relationship between homotopy and languages; [92] provides a starting point.

This WP2.3 will be the focus of the proposed postdoc project at LIX.

#### Deliverables and milestones

| ID   | Deliverable/Milestone                                            | Authors | Date     |
|------|------------------------------------------------------------------|---------|----------|
| D2.1 | Directed homotopy type theory (PhD thesis L.Leclerc)             | LIX     | $T_0+12$ |
| D2.2 | Categories of modules in directed topology (PhD thesis A.Albert) | LIX     | $T_0+24$ |
| D2.3 | Algebraic structure of geometric realizations                    | LIX     | $T_0+48$ |
| D2.4 | Precubical sets and manifolds (PhD thesis Y.Chamoun)             | LIX     | $T_0+12$ |
| D2.5 | Geometric models for concurrent hybrid systems                   | LIX     | $T_0+48$ |
| D2.6 | Homotopy, unfoldings, and computations (Postdoctoral project)    | LMF     | $T_0+36$ |

**Risks and fallbacks.** WP2 is an exploratory workpackage, any results obtained will positively affect the other workpackages, but it bears no risk for the other parts of CHAAT.

#### WP3: Variants and extensions of higher-dimensional automata.

**Workpackage leader:** Amazigh Amrane (LRE)

**Participants:** Bollig, Chatain, Fahrenberg, Faraut (LMF). Amrane, Bazille, Laversa (LRE). Clement, Fortin, Krivine, Ledent (IRIF). Dziadek, Schlehuber-Caissier (SMV).

**Objectives.** There are some variants and generalizations of HDAs which have already shown their usefulness, notably HDAs with interfaces [47, 49], partial HDAs [12, 36], and ST-automata [8, 10, 13]. These have been mostly introduced because standard HDAs do not always suffice in some specific contexts, and other variants may be of use. WP3.1 will develop their classification, which will be used in WP1 and WP2.



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There exist also some *extensions* of HDAs, developed out of interest in specific applications. One of these is real-time concurrency, i.e. systems which combine concurrency and real-time constraints. Most of the current formalisms for modeling real time employ interleaving concurrency, and the standard formalisms for non-interleaving concurrency, including HDAs, use logical instead of real time. For example, there is a natural non-interleaving semantics for Petri nets (WP3.3), but almost all work on real-time extensions of Petri nets, including the popular tool [TAPAAL](#), uses an interleaving semantics. One notable exception to this are *time Petri nets* [88] which do have a non-interleaving real-time semantics [15, 30] where languages are given as sets of pomsets with timestamps on events.

In WP3.2 we will develop the theory of higher-dimensional *timed* automata (HDTAs), a model introduced in [44] and whose language semantics we have developed in [8, 9]. These extend both HDAs and timed automata [4], combining the non-interleaving concurrency of the former with the real-time modeling, using clocks, of the latter. [8, 9, 44] have shown that HDTAs preserve the good properties of timed automata, notably the region construction and zone-based reachability algorithms; but much remains to be explored. We also intend to use HDTAs as operational semantics for timed extensions of Petri nets (WP3.3) and to develop the theory of higher-dimensional *hybrid* automata.

**WP3.1: Variants of HDAs.** There are some generalizations of HDAs which have shown their usefulness: HDAs with interfaces [47, 49] in which some events may not be terminated or others not be unstated; partial HDAs [12, 36] in which some faces may be missing; and ST-automata [8, 10, 13] which are standard automata on an infinite alphabet of starters and terminators and have proven useful as operational semantics for HDAs. Several other generalizations are of possible interest, and we plan to organise them into a classification (M3.1). Most of them also have presheaf characterisations, inducing natural notions of geometric morphisms between them. We will investigate all these variants for their properties and their relationship with one another. The classification will be used in WP1.1, WP1.2 and WP2.4.

There is some indication that partial HDAs are better suited for infinite pomsets, see WP1.2, and that relational HDAs (a to-be-defined generalization of partial HDAs) provide a more natural setting for a Kleene theorem than [47] or a stronger Büchi-Elgot-Trakhtenbrot theorem than [11]. Further, we conjecture that partial HDAs are determinizable, whereas HDAs with interfaces are not [49]. Partial HDAs also naturally appear as operational semantics for extensions of Petri nets [12], making a connection to WP3.3.

Another variant is given by *symmetric* HDAs, i.e. HDAs with actions of the symmetric groups which allow for interchanging directions, which have been used in a context of bisimulation (WP1.3) in [79]. Based on the work in [109], we will integrate them in our classification. For applications in distributed systems, a notion of HDAs enriched with *processes*, i.e. with a map from events to processes which maps concurrent events to different processes, seems important; we will pursue this also in relation to WP3.2.

**WP3.2: Higher-dimensional timed and hybrid automata.** Amrane et al have introduced a new language-based semantics for real-time concurrency in [8, 9], with higher-dimensional *timed* automata (HDTAs) [44] as its operational models. Their semantics is given as languages of pomsets with interval timestamps. As for timed automata, language inclusion for HDTAs is undecidable, whereas it becomes decidable for their untimed languages.

The work in [8, 9] is only a starting point, and HDTAs remain largely unexplored. In particular, we will need to define notions of timed (bi)simulation, guided by the work in WP1.3; we conjecture that these are decidable for HDTAs just like for timed automata. Moreover, incorporating information about event durations and timings raises challenges similar to those encountered in timed automata, which assume unrealistic assumptions such as perfect clock precision and zero-delay actions. With concurrency, it becomes even more relevant to examine the robustness of HDTAs and their languages under delay perturbations. This can be formalized through notions of distance, where timed word distances take symbol permutations into account. Partial orders and interval-timed pomsets appear promising for this purpose. Understanding these robustness properties is crucial for designing more realistic verification algorithms, in connection with WP4.1. This should be achieved using existing timed temporal logics like TCTL [4], or by developing new ones.

[9] also notes some similarities between interval-timed pomsets and multi-dimensional signals [85]. This suggests that distance definitions could benefit from recent work on signal distances [105]. Yet signal semantics are most naturally defined on some notion of HDTAs *with processes* (WP3.1). This opens up new

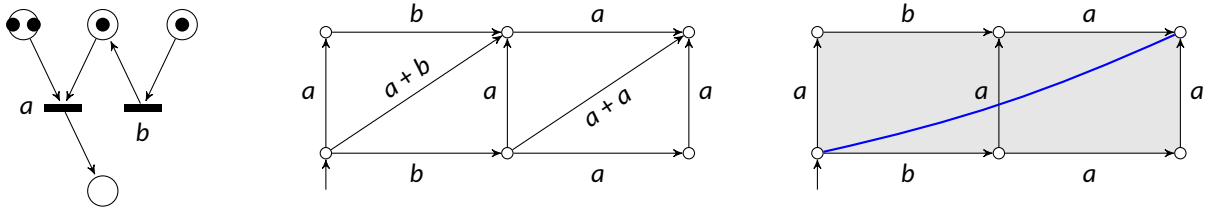


Figure 3: Petri net (left), concurrent step semantics (middle), and HDA semantics (right), including an execution which is not captured by the concurrent step semantics.

avenues, related to WP4.1, for exploring metric (interval) temporal logic (M(I)TL) or signal temporal logic (STL) model checking and synthesis using HDTAs.

In addition to HDTAs, it may be worthwhile to explore generalizations of other models for real-time systems such as event-clock automata [5] or automata with timers [26] which have better algebraic properties. The latter also allows for efficient active learning [25], a connection with WP1.4 which we will explore. Finally, [44] also introduces higher-dimensional *hybrid* automata, i.e. HDAs with real-valued variables which are constrained by differential equations, and [1] gives a new notion of probabilistic concurrent systems which is closely related to HDAs. We will develop their theory, also in relation to WP2.2.

**WP3.3: HDAs as semantics for concurrent formalisms.** HDAs lend themselves well to giving operational semantics to models for concurrent systems. The expressiveness of HDAs supports fine-grained reasoning about concurrent semantics and enables comparisons between different models and their extensions. Moreover, translations to HDAs allow the use of model checking techniques developed in WP4.1.

Van Glabbeek defined translations from Petri nets and event structures to HDAs [112] based on variations of their concurrent semantics. Amrane et al [12] recently revisited the Petri net translations and implemented them in a prototype tool [pn2HDA](#). Compared to concurrent step transition systems [61], HDA semantics is more precise and captures richer concurrent behavior (Fig. 3). [12] also addressed translations from extensions of Petri nets to *partial* HDAs (WP3.1), and we plan to extend these translations and to revisit the translations from event structures to HDAs to refine their concurrent semantics. A concrete application is analyzing Petri net simplification algorithms, e.g., removal of read or inhibitor arcs, with respect to concurrency preservation.

Another model of concurrent systems that allows for non-sequential semantics are communicating automata [24]. In contrast with HDAs, most verification problems for communicating automata are undecidable (although there are positive results regarding *implementability* of certain specifications [21, 22]). Several restrictions that are useful in practice and allow for decidable model-checking have been proposed; for instance synchronizability [81] or realizability with synchronous communication (RSC) [62]. While (unrestricted) communicating automata do not admit an HDA semantics, it would be interesting to compare RSC and other conditions with HDAs and investigate an extension of HDAs with queues for the unrestricted case. Also asynchronous automata [116] and interactions [84] are models that have seen some applications in modeling concurrent systems and should admit some form of HDA semantics which we will investigate. Timed extensions of Petri nets [88, 95] should have a semantics as higher-dimensional *timed* automata (WP3.2) which we will explore.

#### Deliverables and milestones

| ID   | Deliverable/Milestone                        | Authors | Date     |
|------|----------------------------------------------|---------|----------|
| M3.1 | Classification of variants of HDAs           | LIX     | $T_0+24$ |
| D3.2 | Variants of HDAs                             | LIX     | $T_0+36$ |
| D3.3 | Higher-dimensional timed and hybrid automata | SMV     | $T_0+48$ |
| D3.4 | HDAs as semantics for concurrent formalisms  | LRE     | $T_0+48$ |

#### Risks and fallbacks

- **Risk:** In WP3.1, the proposed classification work on HDA variants may be more difficult than planned, delaying M3.1 and impacting other parts of CHAAT. **Risk level:** low. **Impact:** medium. **Fallback:** Even without M3.1, any partial results will be useful for WP1.1, WP1.2 and WP2.4.

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#### WP4: Applications and tools

**Workpackage leader:** Philipp Schlehuber-Caissier (SMV)

**Participants:** Fahrenberg, Faraut, PhD project (LMF). Amrane, Bazille, Pommellet (LRE). Clement, Fortin, Postdoctoral project (IRIF). Goubault, Haucourt, Mimram (LIX). Hajdas, Kushik, Schlehuber-Caissier, PhD project (SMV).

**Objectives.** The purpose of this workpackage is to develop open-source tools and libraries for HDAs that are easy to extend and adapt to specific use-cases, and apply the developed theory to solve concrete control and supervision problems. The goal is therefore threefold. Firstly we hope to show their usefulness and effectiveness for certain modelling and verification tasks well established in the academic community, like the [model checking competition](#). Secondly the creation of such open-source tools should bootstrap a broader adoption in the automata community and possibly beyond. Finally we hope to show its usefulness in an industrial setting, by dealing with problems and situations arising in the context of Industry 4.0 with its high degree of concurrency, possibly even under additional real-time constraints.

These tools will be based on, and will in turn influence the development of, the theory and algorithms derived in other workpackages, specifically those aiming at model checking, translations between models, and learning, mostly located in WP1 and WP3. The workpackage builds on the extensive experience of the partners SMV and LRE in implementation and will allow us to test the developed theory in practice and participate in well established competitions for dissemination. In addition, the LMF partner will provide the necessary expertise in the area of Industry 4.0 and automation to ensure technology transfer to an industrial setting.

**WP4.1: Model checking of HDAs.** This workpackage builds on the results developed in WP1 to enable model checking of HDAs. There are currently very few tools that rely on HDAs as a theoretical foundation: [PG2HDA](#), a tool for converting process graphs to HDAs, and [pn2HDA](#) for translating Petri nets to HDAs. Both tools rely on an explicit representation of the underlying HDA.

That is, each cell and its face-maps are explicitly represented and stored in memory. Similarly, ST-automata, which are an operational model for HDAs that has proven very useful in proofs, use an explicit and overly detailed representation, barring it from being used in practice. However, without losing information, we may also store only the highest-dimensional cells, that is cells that do not appear in the image of a face-map, plus information on relations between different such maximal cells.

We call this the *reduced* or *max-cell* representation and it results in an exponentially smaller memory footprint compared to the explicit representation. First steps in this direction have been taken in [14], however much work remains to be done and further optimizations to be explored. New data structures are required for efficient manipulation and exploration of such automata. Inspiration may be found in the theory of tame paths (WP2.4) and preliminary work indicates strong connections between max-cell HDAs obtained from Petri nets, convex (tropical) polytopes and algebraic topology.

Concerning model checking algorithms as such, there are several sub-directions to explore here. Even for seemingly trivial tasks like acceptance (does a given HDA accept a given pomset) there are currently no efficient algorithms or implementations neither for the reduced nor for the explicit representation. More sophisticated algorithms will be needed for efficient MSO and FO model checking; it should be noted that these have a non-elementary complexity (even over words), but that over words tools such as MONA [20] still perform well in many practical cases.

Another particularly interesting venue is the search for a translation akin to the translation of LTL-formulas to  $\omega$ -automata. That is, a translation from  $LTL_{Poms}$  [32] or similar to  $\omega$ -HDA, opening the door to verification of HDAs. The next step would be reactive synthesis, as usually done for LTL. Here the textbook approach is to translate the formula to a deterministic parity automaton and interpret it as a game arena [102]. If the game can be won, the strategy corresponds to a controller. Adapting this approach to HDAs seems like a promising venue for generating concurrent controllers, see WP4.3.

This workpackage draws inspiration from and will seek to interface with [Spot](#), an open-source project working with  $\omega$ -automata and LTL-formulas developed at LRE [40] to which some of the participants are actively contributing. Furthermore, WP4.1 will constitute the core of the SMV PhD project.

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**WP4.2: Active learning of models for concurrency.** Based on WP1.4, we will design and implement efficient algorithms for learning concurrent models within a dedicated library, interfacing with existing frameworks such as LearnLib. The goal is to extend learning techniques to concurrency-expressing structures.

We will focus on learning algorithms for languages of series-parallel pomsets, interval pomsets (relevant for HDAs), and other concurrency models that can capture a wide range of concurrent systems. This task also overlaps with WP4.1 by contributing to model inference for model-checking, ensuring that learned models can be used for verification purposes.

Ultimately, learning is not just about feasibility but also efficiency. Learning algorithms for words have become increasingly efficient, more general, and less reliant on extraneous information. We aim to achieve the same for concurrency models, developing learning techniques that are scalable and broadly applicable. The interfaces with existing frameworks will allow to evaluate their efficiency on well-known benchmarks set and derived instances intended to highlight the concurrent aspects of the model.

Beyond this purely academic setting, we will seek to apply these techniques in two different contexts. The first one being network protocols and distributed, network based systems and their verification, linking it to WP3.3. Indeed such systems are often complex and their typical specification (for instance in the form of RFCs) allow for some ambiguity, intentionally or not. A common approach is therefore to treat the component in question as a black-box and infer its behavior from observation and interaction for a variety of models with different capabilities as shown in [27, 57, 70]. Using HDAs with their concise representation of concurrency (a cell of dimension  $n$  represents  $n!$  different sequential executions) bears great potential for efficient algorithms. This will constitute the application-oriented part of WP1.4, in which Jakub Hajdas (PhD SMV) and the LMF PhD project will be involved and co-operate on.

The second practical application for HDA learning concerns industrial applications, linking this workpackage also to WP4.3. Here, as physical machines are involved, the paradigm changes from active to passive learning, as we can not simply execute possibly unsafe instructions on the machine. Nonetheless there is a significant overlap concerning base concepts and data structures. The idea is to extract a minimal correct concurrent model for industrial plants, similar to the approach proposed in [108] for extracting interpreted Petri nets.

**WP4.3: HDAs for concurrent controllers** This workpackage is concerned with the application of HDAs in an industrial setting, notably Industry 4.0, to analyse and control production systems showing a high degree of concurrency. It mainly builds upon the theory and tools developed in WP4.1.

A first step in this direction is presented in [18]: There it is shown how to extend the translation from Petri nets to HDAs of [12] to *interpreted* Petri nets [34], i.e. Petri nets showing an input-output behaviour that can be used as a controller. To this end, the authors of [18] introduce interpreted HDAs, a similar extension of HDAs. The translation is then used to expose subtle concurrency errors that are hard to detect when working directly with the interpreted Petri net.

This first publication serves as a proof of concept that HDAs are indeed a good framework to describe concurrency in real-world controllers and opens up a path to further developments. More specifically, several different operational semantics can be associated to interpreted HDAs. We will investigate these different semantics vis-à-vis their impact on the usability and expressiveness of the model. To give a practical example, events in HDAs are not instantaneous, raising the question of what happens if a real-world task is interrupted, i.e. when the termination of an event becomes uncertain. This, in turn, leads to interesting theoretical questions involving a new form of termination non-determinism and opens up connections to relational HDAs (WP3.1). We will also investigate the applicability of our technique to other models known in the community like signal interpreted Petri nets [80].

Another interesting research direction stems from the control cycle time: In an industrial setting the control algorithm is executed on a dedicated embedded system with the read-compute-write cycle performed at a specified rate. Executing this cycle at a higher rate allows a more fine-grained control of the system, as we are sure to observe all intermediate configurations in the correct order, but is also associated to significantly increased hardware cost. Allowing for partial-order semantics here would permit to prove correctness of the controller under less restrictive circumstances, reducing hardware costs.

In a longer-term perspective, also the theory developed in WP3.2 concerning higher-dimensional timed and hybrid automata is of great interest here. Indeed the systems controlled in an industrial setting often

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use physical time, either to represent real-time constraints or to faithfully model the system behavior. The models and algorithms of [8, 9, 44] form a good foundation for this, but they will need to be extended in several directions (input-output behaviour, verification and synthesis algorithms) to be useful in practice.

#### Deliverables and milestones

| ID   | Deliverable/Milestone                                          | Authors | Date               |
|------|----------------------------------------------------------------|---------|--------------------|
| M4.1 | Framework and data structures to represent HDAs                | SMV     | T <sub>0</sub> +18 |
| D4.2 | Model checking of HDAs (PhD thesis)                            | SMV     | T <sub>0</sub> +42 |
| D4.3 | Open-source tool working with HDAs, Petri nets and logics      | SMV     | T <sub>0</sub> +48 |
| D4.4 | Open-source library for learning HDAs and other models         | LRE     | T <sub>0</sub> +48 |
| D4.5 | Concurrent control design via HDAs                             | SMV     | T <sub>0</sub> +24 |
| D4.6 | Verification and synthesis of real-time concurrent controllers | LMF     | T <sub>0</sub> +48 |

#### Risks and fallbacks

- **Risk:** Our proposed data structures and algorithms for the reduced representation of HDAs (D4.1) may be more difficult to devise than thought. **Risk level:** medium. **Impact:** low. **Fallback:** Between the explicit and reduced representations there are many partial optimizations which may be done, and all will improve our model checking tool.

#### WP5: Project management and dissemination.

**Workpackage leader:** Uli Fahrenberg (LMF)

**Participants:** all

**Objectives.** WP5 has a twofold objective: project management for ensuring the success of CHAAT and dissemination of the contributions and tools produced by the project as outlined in Fig. 4.

Project management includes coordination and monitoring, delivering the expected results on time, budget follow-up, website and repository setup and maintenance, meeting organization, and communication. The project coordination will rely on regular communication actions: a kick-off meeting; face-to-face project meetings to discuss progress and assess milestones, held bi-annually and alternating between partners; technical meetings organized by WP leaders to address specific tasks.

We will set up a **Project Management Committee**, consisting of the project coordinator, the partner coordinators and the WP leaders, which will meet online at least every 3 months in order to coordinate the project and ensure that frequent exchanges between partners will be done. In addition to the Project Management Committee we will install an international **Scientific Advisory Board** to guide and advance the project's research and to assure its impact, see Introduction.

Regarding **dissemination**, a public website will be set up and regularly maintained to advertise results of the project. A companion private website will be used for internal communication. All project contributions will be managed at a centralized git repository hosted by SMV. We will attend and present our results at the main conferences in our domain, i.e. events related to automata and concurrency theory and its applications, e.g.

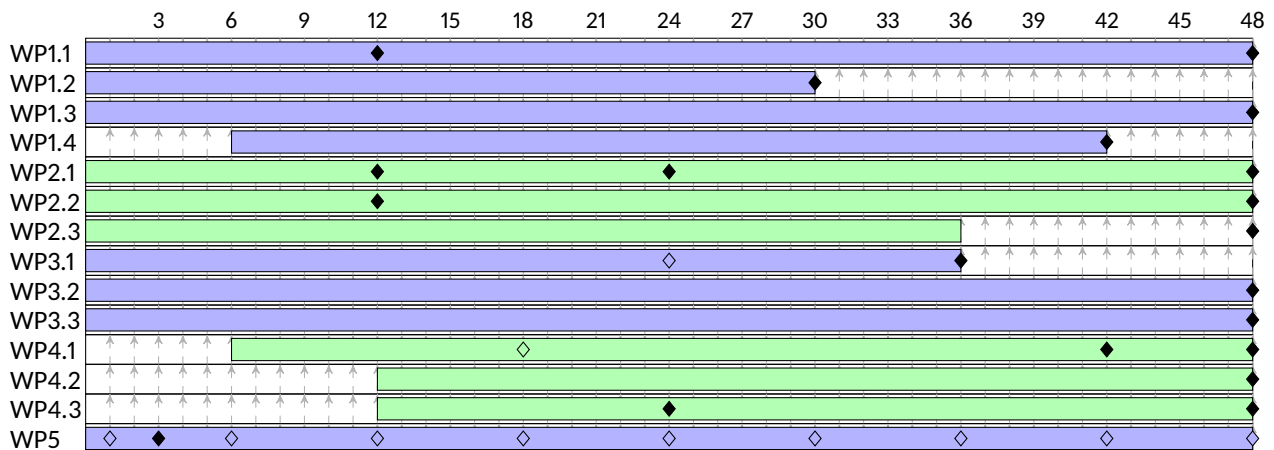


Figure 4: Gantt diagram of CHAAT. Black diamonds denote deliverables, lozenges milestones



|                                    |                |           |            |
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CONCUR, ETAPS, LICS, ICALP, STACS, PETRI NETS, CAV. We will make our software publicly available. We will organize a public workshop at the end of the project, tentatively at CIRM and organized by the LMF partner, to present the results of CHAAT.

The project partners are jointly organizing the [Paris Automata and Concurrency Theory Seminar](#), created to reinforce the links between different research groups in the Paris area which are working on the connections between automata and concurrency theory. We will use this venue to present and advance our research. Partners LMF and LIX are involved in the Steering Committees of the international conferences [GETCO](#) and [RAMiCS](#), and we ask for funding to organize an edition of each in France during the project.

#### Deliverables and milestones

| ID   | Deliverable/Milestone                   | Authors                      | Date                                                    |
|------|-----------------------------------------|------------------------------|---------------------------------------------------------|
| M5.1 | Kick-off meeting                        | LMF                          | $T_0+1$                                                 |
| D5.2 | Web site                                | LMF                          | $T_0+3$ ; continuous update                             |
| D5.3 | Git repository (open-source code)       | SMV                          | $T_0+3$ ; continuous update                             |
| M5.4 | Biannual project meeting                | alternating between partners | $T_0+6, T_0+12, T_0+18, T_0+24, T_0+30, T_0+36, T_0+42$ |
| M5.5 | Final project meeting & public workshop | LMF                          | $T_0+48$                                                |
| M5.6 | Coordination meeting (online)           | All coordinators             | every 3 months                                          |

#### Risks and fallbacks

- **Risk:** We may encounter difficulties in the recruitment of PhD students and postdocs due to the state of the job market in Computer Science. **Risk level:** **high**. **Impact:** **medium**. **Fallback:** While parts of the project rely on these recruitments, the four-year spread of the project allows for much flexibility.

## II. Organisation and implementation of the project

### a. Partnership (consortium or team)

**Scientific coordinator.** [Uli Fahrenberg](#) is professor at LMF and Université Paris-Saclay and will devote 70% of his research time to the CHAAT project. He is an experienced researcher specializing in real-time systems, quantitative verification, directed algebraic topology, and concurrency theory [42, 43]. He has initiated the recent wave of research in HDAs and their extensions [8, 11, 45, 47, 49] and succeeded in creating a community dedicated to HDA research, most of whom are participating in this proposal. He also has project managing experience, having acted as task manager for several ANR and European projects. Fahrenberg is coordinator of an application for a CNRS International Research Project involving Dubut, Ledent, Struth, and Ziemiański and related to this proposal.

**Consortium and its complementarity.** CHAAT gathers a community of researchers interested in HDAs and other models for concurrency from many different viewpoints, and the project will be a driving force in promoting collaboration between these partners. The core of the project consists of the scientific managers of the partners LMF, IRIF, LIX and LRE, all of which have rich experience in concurrency models in general and in the theory and applications of HDAs in particular [7, 35, 42, 83], and of the manager of the SMV partner who is responsible for applications and tools. Together, the partners animate the [Paris Automata and Concurrency Theory Seminar](#), linking different research groups in the Paris area which are working on automata theory and concurrency theory. The various subgroups involved in this project have rather different backgrounds and experience, so they are likely to complement each other in driving the project.

**LMF: Laboratoire méthodes formelles.** The LMF partner involves four members of its Modeling cluster and one member of the laboratoire LURPA and is headed by the CHAAT coordinator Uli Fahrenberg. It is involved in WP1, WP2, WP3, and WP4 and brings experience in concurrency theory, HDAs, and Petri nets which is essential for the project. LMF asks for funding for one PhD student who will work on WP1 together with IRIF.

**IRIF: Institut de recherche en informatique fondamentale.** The IRIF partner involves three members of its Algebra and Automata research groups and one member associated from LIPN and is headed by Jérémy Ledent, Assistant Professor at Paris Cité University and an expert in concurrent and distributed systems. It will lead WP1 and is involved in WP3 and brings experience in concurrency theory and algebra which is necessary for the project to succeed. IRIF asks funding for a postdoc to support its work on WP1.

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| <b>AAPG2026</b>                    | <b>CHAAT</b>   |           | <b>PRC</b> |
| Coordinated by:                    | Uli FAHRENBURG | 48 months | 672k€      |
| Axe E.01 : Fondements du numérique |                |           |            |

**LIX: Laboratoire d'informatique de l'École polytechnique.** The LIX partner comprises five members of its Cosynus and Partout research groups and is headed by Jérémy Dubut, Assistant Professor at École polytechnique and an expert in concurrency, HDAs, and bisimulations. It will lead WP2 and is involved in WP1 and WP4 and brings expertise in geometry and topology and their application to concurrency theory which is essential for the project. LIX asks for funding for a postdoc to support its work on WP2.

**LRE: Laboratoire de recherche de l'EPITA.** EPITA is a private software engineering university with a diploma recognized by the French CTI, and its research lab LRE is evaluated by the Hcéres. The LRE partner involves four members of its Automata research group and is headed by Amazigh Amrane, Assistant Professor at EPITA and an expert in HDAs and pomsets. It will lead WP3 and is involved in WP1 and WP4 and brings experience in HDAs and algorithms which is necessary for the project. LRE is a full-cost beneficiary.

**SMV: Services répartis, architectures, modélisation, validation, administration des réseaux.** The SMV partner involves three members of its NeSS and ACMES research groups and is headed by Philipp Schlehuber-Caissier, Assistant Professor at Télécom SudParis and an expert in algorithms and automata theory. It will lead WP4 and is involved in WP3 and brings expertise in algorithms and implementation which is essential for the project to succeed. SMV asks for funding for one PhD student who will work on WP4 together with LRE.

**PhD and postdoc projects.** The CHAAT partners are asking for funding for 2 PhD projects and 2 postdocs, also mentioned above:

- PhD project at LMF, co-supervised by IRIF, on WP1.4 and WP4.2. This thesis aims to design learning algorithms for concurrent systems, with a focus on HDAs and their variants from WP3, while also exploring other concurrency models. The goal is to extend automata learning techniques to settings that capture causality, concurrency, and non-interleaving behaviors. In particular, the work will investigate the learnability of HDAs and related models, both from an algorithmic and a theoretical perspective, by studying structural properties that impact inference.  
Beyond theory, the thesis also targets practical applications. This includes exploring connections with model-based testing and related techniques, in order to enable the integration of learning methods into broader verification and analysis frameworks. The developed algorithms will be implemented in a dedicated library and integrated with existing tools (D4.4), enabling automated analysis and model inference for real-world concurrent systems. The full description of the PhD proposal can be found [here](#).
- PhD project at SMV, co-supervised by LRE, working on WP4.1 and WP4.3 with links to WP1.1 and WP1.2. The goal is to derive efficient model checking algorithms based on the language theory developed in WP1.1 and WP1.2. To this end the PhD candidate will work closely with the participants of WP1.1 and WP1.2 not only to thoroughly understand the theory, but also to guide its development based on needs of model checking. The PhD candidate will play an important role in delivering the implementations and tools in D4.1 together with partners from SMV and LRE. Moreover the candidate is invited to contribute to real-world applications in D4.2 together with partners from LMF. The full description of the PhD proposal can be found [here](#).
- postdoc project at LIX, in collaboration with LMF, on WP2.3. The goal of this project is to develop a proper notion of unfolding of HDAs. This requires classifying which variants of HDAs (e.g., partial or not) and which type of paths (e.g., tame or not, modulo homotopy or not) are relevant. Particular attention should be given to how this construction relates to the unfolding of Petri nets, including which computational methods can be carried over to HDAs, as well as to the universal covering of directed spaces, highlighting the interplay between homotopy and concurrency.
- postdoc project at IRIF, in collaboration with SMV, on WP1.2 and WP4.1. The goal is to develop translations between LTL-type logics for HDAs (part of Erlich's ongoing PhD) and  $\omega$ -HDAs, inspired by the translations for standard LTL and  $\omega$ -automata [75, 102] used for example in [Spot](#). This requires developing notions of alternating HDAs and parity acceptance in WP1.2 (part of Hajdas' ongoing PhD) but, more importantly, algorithms for translations. These algorithms will be implemented in WP4.1 as part of our model-checking effort.

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**Gender and age balance.** Only about 19% (4 out of 21) of the permanent staff involved in CHAAT are women, which is below the average for CNRS section 2 (22%). We will attempt to achieve better gender balance when hiring interns, PhD students and postdocs for the project.

CHAAT has a significant involvement of early-career researchers, also as WP leaders, and the project is expected to promote their careers: about 76% (16 out of 21) of the permanent staff in CHAAT are of rank B (associate professor or researcher), above the average for CNRS section 2 (52%).

**Implication of the scientific coordinator and the scientific leaders of partners in on-going project(s):**

| Name   | p.m. | Call, funding agency, grant | Project title | Coordinator | Start - end |
|--------|------|-----------------------------|---------------|-------------|-------------|
| Ledent | 6    | ANR-23-CE48-0005            | PaVeDys       | R.Iosif     | 2024-27     |

**b. Implemented and requested resources to reach the objectives**

**Partner 1: LMF: Laboratoire méthodes formelles**

**Staff expenses.** LMF will invest 62.4p.m. of permanent staff in CHAAT. 1 PhD student, co-supervised by LMF and IRIF, to work on WP1.4, 36p.m. (128k€), and 1 internship, 6p.m. (4.2k€).

**Equipments.** 1 laptop for PhD student (2k€), 2k€ books etc.

**Other costs.** Travel cost 16.5k€ (project meetings; dissemination).

**Partner 2: LRE: Laboratoire de recherche de l'EPITA**

**Staff expenses.** LRE is a full-cost beneficiary and demands funding for 52.8p.m. of permanent staff plus 7.2p.m. of support staff (310k€). Additionally, LRE invests 6p.m. of non-permanent staff in CHAAT. 2 internships, 12p.m. (9k€).

**Equipments.** 2k€ books etc.

**Other costs.** Travel costs 14.5k€ (project meetings; dissemination).

*Taux d'aide : 25%*

**Partner 3: IRIF: Institut de recherche en informatique fondamentale**

**Staff expenses.** IRIF will invest 43.2p.m. of permanent staff in CHAAT. 1 postdoc, to work on WP1, 12p.m. (66.7k€), and 2 internships, 12p.m. (8k€).

**Equipments.** 1 laptop for postdoc (2k€), 1.2k€ books etc.

**Other costs.** Travel cost 11.5k€ (project meetings; dissemination).

**Partner 4: LIX: Laboratoire d'informatique de l'École polytechnique**

**Staff expenses.** LIX will invest 74.4p.m of permanent staff in CHAAT and 24p.m of non-permanent staff. 1 postdoc, to work on WP2, 12p.m (67.1k€).

**Equipments.** 1 laptop for postdoc (2k€), 3.2k€ books etc.

**Other costs.** Travel cost 24k€ (project meetings; dissemination).

**Partner 5: SMV: Télécom SudParis, SAMOVAR**

**Staff expenses.** SMV will invest 53.6p.m. of permanent staff in CHAAT. 1 PhD student, co-supervised by SMV and LRE, to work on WP4.1 and WP4.3, 36p.m. (145k€). 1 internship, 6p.m. (4.5k€), to help with WP4.

**Equipments.** 1 laptop for the PhD student (2k€)

**Other costs.** Travel costs 11k€ (project meetings; dissemination), 1.6k€ books etc.

**Requested means by item of expenditure and by partner (all in €)**

|                                             | LMF           | IRIF   | LIX    | LRE   | SMV    |
|---------------------------------------------|---------------|--------|--------|-------|--------|
| Staff expenses                              | 132162        | 74700  | 67100  | 79795 | 149500 |
| Equipments                                  | 4400          | 3200   | 5200   | 500   | 4000   |
| Other costs                                 | 19500         | 11500  | 24000  | 3625  | 12500  |
| Administrative management & structure costs | 21068         | 12069  | 13001  | 12258 | 22410  |
| <b>Sub-total</b>                            | 177130        | 101469 | 109301 | 96178 | 188410 |
| <b>Requested funding</b>                    | <b>672488</b> |        |        |       |        |

LRE is a full-cost beneficiary; in the table, a *taux d'aide* of 25% has been applied.

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### III. Impact and benefits of the project

Given that CHAAT is a fundamental research project, we will measure its success by the influence of our work on the involved research communities. Results will be disseminated through presentations at relevant conferences, e.g. CONCUR, ETAPS, LICS, ICALP, STACS, PETRI NETS, CAV, and through journal publications. Further, the planned self-contained monograph on the state of the art on HDAs will hopefully serve as a reference on the subject for years to come.

We will also make our code publicly available in order to contribute to the state of the art in model checking, active learning, and industrial control. The CHAAT project has the potential to have a significant impact in the model checking of concurrent systems, given that higher-dimensional automata faithfully model concurrency and their reduced representation is exponentially smaller than other representations.

The potential societal benefits brought about by the CHAAT project are numerous. Model checking helps ensure the correctness, reliability, and safety of critical systems that we all depend on, and it is important to better tame the concurrency inherent in most critical systems. Similarly, the control of concurrent systems is essential for Industry 4.0. More generally, a wide range of application domains, where the system dynamics display control, synchronization, communication, or decision making, will all benefit of a better understanding and mastering of their inherent concurrency features, which in turn will lead to better tools for their analysis and design.

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| Axe E.01 : Fondements du numérique |                |           |            |

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| Axe E.01 : Fondements du numérique |                |           |            |

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